

**Table 1:** Single enzyme — NRMSE (median).

| (a) Full observation |                   |                   |                   |                   | (b) First-last observation |                   |                   |                   |                   |
|----------------------|-------------------|-------------------|-------------------|-------------------|----------------------------|-------------------|-------------------|-------------------|-------------------|
| Model                | $n = 3$           | $n = 10$          | $n = 100$         | $n = 1000$        | Model                      | $n = 3$           | $n = 10$          | $n = 100$         | $n = 1000$        |
| <b>Enzyme/2</b>      |                   |                   |                   |                   | <b>Enzyme/4</b>            |                   |                   |                   |                   |
| CMVF                 | 0.526             | 0.148             | 0.044             | 0.012             | CMVF                       | <b>0.030</b>      | 0.060             | <b>0.004</b>      | <b>0.003</b>      |
| CMVF-L1              | 0.526             | 0.148             | <b>0.042</b>      | 0.014             | CMVF-L1                    | 0.038             | 0.060             | 0.008             | 0.007             |
| CMVF-unb.            | 0.606             | <b>0.080</b>      | 0.046             | <b>0.009</b>      | CMVF-unb.                  | 0.037             | <b>0.024</b>      | 0.006             | $1.0 \times 10^6$ |
| NODE-GRU             | 0.776             | 0.103             | 0.077             | 0.015             | NODE-GRU                   | 0.520             | 0.478             | 0.253             | 0.232             |
| NODE-MLP             | 0.729             | 0.116             | 0.180             | 0.222             | NODE-MLP                   | 0.279             | 0.193             | 0.237             | 0.263             |
| NODE-corr.           | <b>0.424</b>      | 0.083             | 0.171             | 0.219             | NODE-corr.                 | 0.067             | 0.050             | 0.094             | 0.063             |
| Global ( $\theta$ )  | 0.466             | 0.348             | 0.208             | 0.253             | Global ( $\theta$ )        | 0.031             | 0.031             | 0.035             | 0.041             |
| IC ( $\theta$ )      | 0.500             | 0.180             | 0.211             | 0.244             | IC ( $\theta$ )            | 0.031             | 0.028             | 0.035             | $1.0 \times 10^6$ |
| <b>Enzyme/4</b>      |                   |                   |                   |                   | <b>Enzyme/6</b>            |                   |                   |                   |                   |
| CMVF                 | 0.031             | 0.074             | <b>0.004</b>      | <b>0.003</b>      | CMVF                       | 0.012             | 0.009             | <b>0.002</b>      | <b>0.002</b>      |
| CMVF-L1              | 0.038             | 0.041             | 0.009             | 0.006             | CMVF-L1                    | 0.010             | 0.010             | 0.004             | 0.005             |
| CMVF-unb.            | 0.046             | 0.032             | 0.015             | 0.003             | CMVF-unb.                  | 0.012             | 0.005             | 0.004             | 0.002             |
| NODE-GRU             | 0.733             | 0.139             | 0.020             | 0.008             | NODE-GRU                   | $5.4 \times 10^7$ | $6.6 \times 10^7$ | $1.6 \times 10^7$ | $1.6 \times 10^7$ |
| NODE-MLP             | 0.509             | 0.184             | 0.028             | 0.005             | NODE-MLP                   | $1.4 \times 10^7$ | $5.0 \times 10^7$ | $1.5 \times 10^7$ | $1.6 \times 10^7$ |
| NODE-corr.           | 0.039             | 0.031             | 0.033             | $1.0 \times 10^6$ | NODE-corr.                 | $7.7 \times 10^6$ | $2.4 \times 10^7$ | $1.9 \times 10^7$ | $1.4 \times 10^7$ |
| Global ( $\theta$ )  | 0.031             | 0.030             | 0.036             | 0.041             | Global ( $\theta$ )        | 0.020             | 0.021             | 0.014             | 0.007             |
| IC ( $\theta$ )      | <b>0.030</b>      | <b>0.028</b>      | 0.034             | $1.0 \times 10^6$ | IC ( $\theta$ )            | <b>0.009</b>      | <b>0.002</b>      | 0.003             | 0.003             |
| <b>Enzyme/6</b>      |                   |                   |                   |                   |                            |                   |                   |                   |                   |
| CMVF                 | 0.013             | 0.007             | <b>0.002</b>      | <b>0.002</b>      |                            |                   |                   |                   |                   |
| CMVF-L1              | <b>0.007</b>      | 0.010             | 0.008             | 0.004             |                            |                   |                   |                   |                   |
| CMVF-unb.            | 0.011             | 0.005             | 0.004             | 0.002             |                            |                   |                   |                   |                   |
| NODE-GRU             | $4.3 \times 10^6$ | $1.9 \times 10^6$ | $7.0 \times 10^5$ | $1.4 \times 10^5$ |                            |                   |                   |                   |                   |
| NODE-MLP             | $7.2 \times 10^6$ | $1.0 \times 10^6$ | $4.0 \times 10^5$ | $5.8 \times 10^4$ |                            |                   |                   |                   |                   |
| NODE-corr.           | $3.0 \times 10^6$ | $2.4 \times 10^5$ | $8.7 \times 10^4$ | $1.2 \times 10^4$ |                            |                   |                   |                   |                   |
| Global ( $\theta$ )  | 0.019             | 0.014             | 0.013             | 0.007             |                            |                   |                   |                   |                   |
| IC ( $\theta$ )      | 0.009             | <b>0.002</b>      | 0.003             | 0.003             |                            |                   |                   |                   |                   |

**Table 2:** MOF synthesis — NRMSE (median).

| (a) Full observation |              |              |              |              | (b) First-last observation |              |              |              |              |
|----------------------|--------------|--------------|--------------|--------------|----------------------------|--------------|--------------|--------------|--------------|
| Model                | $n = 3$      | $n = 10$     | $n = 100$    | $n = 1000$   | Model                      | $n = 3$      | $n = 10$     | $n = 100$    | $n = 1000$   |
| <b>MOF/4</b>         |              |              |              |              | <b>MOF/4</b>               |              |              |              |              |
| CMVF                 | 0.181        | 0.155        | <b>0.107</b> | <b>0.035</b> | CMVF                       | 1.050        | 0.540        | <b>0.443</b> | 0.424        |
| CMVF-L1              | 0.210        | 0.147        | 0.113        | 0.045        | CMVF-L1                    | 1.043        | 0.602        | 0.491        | 0.461        |
| CMVF-unb.            | 0.534        | 0.133        | 0.135        | 0.043        | CMVF-unb.                  | <b>0.735</b> | <b>0.424</b> | 0.605        | <b>0.115</b> |
| NODE-GRU             | 0.407        | 0.159        | 0.120        | 0.051        | NODE-GRU                   | 1.989        | 3.658        | 9.673        | 3.283        |
| NODE-MLP             | 1.045        | 0.132        | 0.204        | 0.196        | NODE-MLP                   | 0.871        | 3.890        | 3.127        | 1.753        |
| NODE-corr.           | 0.401        | <b>0.131</b> | 0.151        | —            | NODE-corr.                 | 0.979        | 0.865        | 0.487        | 0.631        |
| Global ( $\theta$ )  | 0.593        | 0.501        | 0.313        | 0.145        | Global ( $\theta$ )        | 0.948        | 0.668        | 0.743        | 0.887        |
| IC ( $\theta$ )      | <b>0.178</b> | 0.159        | 0.115        | 0.149        | IC ( $\theta$ )            | 0.743        | 0.571        | 0.703        | 0.840        |
| <b>MOF/6</b>         |              |              |              |              | <b>MOF/6</b>               |              |              |              |              |
| CMVF                 | 0.156        | <b>0.067</b> | <b>0.054</b> | <b>0.030</b> | CMVF                       | 0.338        | <b>0.130</b> | 0.120        | 0.092        |
| CMVF-L1              | 0.250        | 0.075        | 0.057        | 0.045        | CMVF-L1                    | 0.273        | 0.144        | <b>0.076</b> | <b>0.080</b> |
| CMVF-unb.            | 0.196        | 0.079        | 0.064        | 0.034        | CMVF-unb.                  | 0.264        | 0.132        | 0.082        | 0.094        |
| NODE-GRU             | 0.393        | 0.181        | 0.159        | 0.053        | NODE-GRU                   | 1.540        | 1.173        | 2.057        | 3.051        |
| NODE-MLP             | 0.408        | 0.182        | 0.135        | 0.084        | NODE-MLP                   | 5.559        | 16.193       | 11.289       | 3.485        |
| NODE-corr.           | 0.317        | 0.140        | 0.069        | 0.060        | NODE-corr.                 | 0.328        | 0.309        | 0.206        | 0.265        |
| Global ( $\theta$ )  | 0.240        | 0.255        | 0.096        | 0.095        | Global ( $\theta$ )        | <b>0.229</b> | 0.240        | 0.116        | 0.178        |
| IC ( $\theta$ )      | <b>0.111</b> | 0.154        | 0.076        | 0.094        | IC ( $\theta$ )            | 0.265        | 0.172        | 0.184        | 0.159        |
| <b>MOF/8</b>         |              |              |              |              | <b>MOF/8</b>               |              |              |              |              |
| CMVF                 | 0.215        | <b>0.052</b> | <b>0.019</b> | <b>0.011</b> | CMVF                       | 0.265        | <b>0.151</b> | <b>0.060</b> | <b>0.035</b> |
| CMVF-L1              | 0.303        | 0.114        | 0.026        | 0.018        | CMVF-L1                    | 0.289        | 0.156        | 0.092        | 0.067        |
| CMVF-unb.            | <b>0.200</b> | 0.118        | 0.026        | 0.015        | CMVF-unb.                  | 0.268        | 0.152        | 0.111        | 0.043        |
| NODE-GRU             | 0.370        | 0.246        | 0.117        | 0.054        | NODE-GRU                   | 3.369        | 3.087        | 44.252       | 22.203       |
| NODE-MLP             | 0.388        | 0.216        | 0.188        | 0.067        | NODE-MLP                   | 3.183        | 10.429       | 9.217        | 7.295        |
| NODE-corr.           | 0.346        | 0.173        | 0.078        | 0.045        | NODE-corr.                 | 0.409        | 0.332        | 0.267        | 0.658        |
| Global ( $\theta$ )  | 0.300        | 0.265        | 0.164        | 0.084        | Global ( $\theta$ )        | 0.272        | 0.259        | 0.178        | 0.117        |
| IC ( $\theta$ )      | 0.249        | 0.130        | 0.072        | 0.082        | IC ( $\theta$ )            | <b>0.240</b> | 0.160        | 0.193        | 0.177        |
| <b>MOF/12</b>        |              |              |              |              | <b>MOF/12</b>              |              |              |              |              |
| CMVF                 | <b>0.086</b> | <b>0.014</b> | <b>0.008</b> | 0.007        | CMVF                       | <b>0.119</b> | 0.112        | 0.044        | <b>0.011</b> |
| CMVF-L1              | 0.167        | 0.027        | 0.020        | 0.009        | CMVF-L1                    | 0.119        | 0.113        | 0.050        | 0.019        |
| CMVF-unb.            | 0.122        | 0.021        | 0.017        | <b>0.007</b> | CMVF-unb.                  | 0.173        | <b>0.080</b> | 0.039        | 0.028        |
| NODE-GRU             | 0.450        | 0.339        | 0.133        | 0.047        | NODE-GRU                   | 3.777        | 3.715        | 3.643        | 3.855        |
| NODE-MLP             | 0.543        | 0.272        | 0.130        | 0.040        | NODE-MLP                   | 1.358        | 2.121        | 2.438        | 1.765        |
| NODE-corr.           | 0.200        | 0.111        | 0.055        | 0.015        | NODE-corr.                 | 0.185        | 0.314        | 0.185        | 0.181        |
| Global ( $\theta$ )  | 0.193        | 0.160        | 0.080        | 0.016        | Global ( $\theta$ )        | 0.169        | 0.167        | 0.127        | 0.051        |
| IC ( $\theta$ )      | 0.094        | 0.025        | 0.010        | 0.009        | IC ( $\theta$ )            | 0.142        | 0.141        | <b>0.027</b> | 0.024        |

**Table 3:** Glycolysis — NRMSE (median).

| (a) Full observation |              |                   |                   |                   | (b) First-last observation |                   |                   |                   |                   |
|----------------------|--------------|-------------------|-------------------|-------------------|----------------------------|-------------------|-------------------|-------------------|-------------------|
| Model                | $n = 3$      | $n = 10$          | $n = 100$         | $n = 1000$        | Model                      | $n = 3$           | $n = 10$          | $n = 100$         | $n = 1000$        |
| <b>Glyc/4</b>        |              |                   |                   |                   | <b>Glyc/4</b>              |                   |                   |                   |                   |
| CMVF                 | <b>0.234</b> | 0.226             | 0.099             | <b>0.056</b>      | CMVF                       | 0.493             | 1.247             | 1.757             | 3.994             |
| CMVF-L1              | 0.236        | 0.253             | 0.124             | 0.061             | CMVF-L1                    | 0.347             | 0.859             | 0.991             | 1.299             |
| CMVF-unb.            | 0.261        | 0.545             | 0.209             | 0.066             | CMVF-unb.                  | 2.589             | 2.058             | 1.370             | 26.702            |
| NODE-GRU             | 0.239        | 0.165             | 0.102             | 0.072             | NODE-GRU                   | <b>0.329</b>      | <b>0.438</b>      | <b>0.360</b>      | <b>0.244</b>      |
| NODE-MLP             | 0.268        | 0.141             | 0.118             | 0.103             | NODE-MLP                   | 0.532             | 0.572             | 0.471             | 6.843             |
| NODE-corr.           | 0.295        | <b>0.100</b>      | <b>0.092</b>      | 0.098             | NODE-corr.                 | 0.378             | 1.114             | 0.999             | 0.318             |
| Global ( $\theta$ )  | 0.480        | 0.876             | 0.494             | 0.221             | Global ( $\theta$ )        | 0.565             | 1.062             | 0.817             | 0.758             |
| IC ( $\theta$ )      | 0.267        | 0.225             | 0.183             | 0.216             | IC ( $\theta$ )            | 0.523             | 0.775             | 0.931             | 0.791             |
| <b>Glyc/8</b>        |              |                   |                   |                   | <b>Glyc/8</b>              |                   |                   |                   |                   |
| CMVF                 | <b>0.146</b> | 0.112             | 0.102             | 0.090             | CMVF                       | 0.250             | 0.485             | 0.679             | 0.867             |
| CMVF-L1              | 0.181        | 0.146             | 0.138             | 0.079             | CMVF-L1                    | 0.279             | 0.772             | 0.704             | 1.292             |
| CMVF-unb.            | 0.192        | <b>0.078</b>      | <b>0.099</b>      | <b>0.062</b>      | CMVF-unb.                  | 0.251             | 0.593             | <b>0.639</b>      | 0.985             |
| NODE-GRU             | 0.169        | 0.135             | 0.117             | 0.102             | NODE-GRU                   | 0.377             | 0.419             | 0.769             | <b>0.617</b>      |
| NODE-MLP             | 0.271        | 0.112             | 0.108             | 0.093             | NODE-MLP                   | 0.311             | <b>0.378</b>      | 1.812             | 2.911             |
| NODE-corr.           | 0.247        | 0.169             | 0.136             | 0.119             | NODE-corr.                 | 0.902             | 2.396             | 1.246             | 5.578             |
| Global ( $\theta$ )  | 0.569        | 1.015             | 0.712             | 0.266             | Global ( $\theta$ )        | 0.564             | 1.013             | 0.653             | 1.437             |
| IC ( $\theta$ )      | 0.182        | 0.442             | 0.175             | 0.202             | IC ( $\theta$ )            | <b>0.161</b>      | 0.461             | 2.949             | 3.646             |
| <b>Glyc/12</b>       |              |                   |                   |                   | <b>Glyc/12</b>             |                   |                   |                   |                   |
| CMVF                 | 0.610        | 0.266             | <b>0.132</b>      | <b>0.065</b>      | CMVF                       | 0.309             | 0.376             | 0.564             | 0.455             |
| CMVF-L1              | 0.259        | 0.308             | 0.148             | 0.068             | CMVF-L1                    | 0.307             | 0.323             | 0.469             | $7.8 \times 10^6$ |
| CMVF-unb.            | <b>0.215</b> | 0.266             | 0.161             | 0.077             | CMVF-unb.                  | <b>0.300</b>      | 0.644             | 0.617             | <b>0.249</b>      |
| NODE-GRU             | 0.282        | 0.314             | 0.157             | 0.123             | NODE-GRU                   | 1.627             | 2.464             | 1.307             | 5.224             |
| NODE-MLP             | 0.252        | 0.312             | 0.188             | 0.116             | NODE-MLP                   | 2.120             | 2.667             | 2.999             | 73.126            |
| NODE-corr.           | 0.266        | 0.266             | 0.152             | 0.149             | NODE-corr.                 | 2.021             | 2.678             | 0.987             | 4.664             |
| Global ( $\theta$ )  | 0.334        | 0.609             | 0.425             | 0.160             | Global ( $\theta$ )        | 0.333             | 0.553             | <b>0.348</b>      | 0.356             |
| IC ( $\theta$ )      | 0.253        | <b>0.211</b>      | 0.151             | 0.162             | IC ( $\theta$ )            | 0.353             | <b>0.272</b>      | 0.522             | 0.654             |
| <b>Glyc/22</b>       |              |                   |                   |                   | <b>Glyc/22</b>             |                   |                   |                   |                   |
| CMVF                 | 0.375        | 0.069             | 0.020             | $6.5 \times 10^6$ | CMVF                       | 0.638             | 0.273             | <b>0.167</b>      | 0.206             |
| CMVF-L1              | 0.812        | 0.069             | 0.025             | 0.005             | CMVF-L1                    | 0.634             | 0.387             | 0.202             | 0.236             |
| CMVF-unb.            | <b>0.095</b> | <b>0.041</b>      | <b>0.015</b>      | <b>0.003</b>      | CMVF-unb.                  | 0.564             | 0.423             | 0.280             | <b>0.103</b>      |
| NODE-GRU             | 562.012      | 663.087           | 506.099           | 0.130             | NODE-GRU                   | $6.7 \times 10^6$ | $1.4 \times 10^7$ | $1.4 \times 10^7$ | 3.007             |
| NODE-MLP             | 2.337        | $3.1 \times 10^5$ | 268.049           | 0.056             | NODE-MLP                   | $7.0 \times 10^6$ | $3.0 \times 10^6$ | $1.4 \times 10^6$ | 35.425            |
| NODE-corr.           | 0.285        | 0.231             | $1.5 \times 10^3$ | —                 | NODE-corr.                 | 0.436             | 1.768             | $5.4 \times 10^6$ | —                 |
| Global ( $\theta$ )  | 0.397        | 0.343             | 0.255             | 0.037             | Global ( $\theta$ )        | <b>0.434</b>      | 0.388             | 0.363             | —                 |
| IC ( $\theta$ )      | 0.310        | 0.112             | 0.027             | 0.012             | IC ( $\theta$ )            | 0.809             | <b>0.270</b>      | 0.271             | —                 |

**Table 4:** Architecture sweep — single enzyme — NRMSE (median).

| Model           | $n = 3$      | $n = 10$     | $n = 100$    | $n = 1000$   |
|-----------------|--------------|--------------|--------------|--------------|
| <b>Enzyme/2</b> |              |              |              |              |
| GRU             | —            | —            | 0.044        | —            |
| LSTM            | —            | —            | 0.065        | —            |
| sLSTM           | <b>0.533</b> | <b>0.114</b> | 0.055        | <b>0.009</b> |
| Transformer     | —            | —            | <b>0.033</b> | —            |
| Mamba           | —            | —            | 0.120        | —            |

**Table 5:** Architecture sweep — MOF synthesis — NRMSE (median).

| Model                     | $n = 3$      | $n = 10$     | $n = 100$    | $n = 1000$        |
|---------------------------|--------------|--------------|--------------|-------------------|
| <b>MOF/4 — full obs</b>   |              |              |              |                   |
| GRU                       | 0.181        | 0.155        | 0.107        | <b>0.035</b>      |
| LSTM                      | 0.163        | 0.151        | 0.115        | 0.041             |
| sLSTM                     | 0.188        | 0.155        | 0.118        | 0.036             |
| Transformer               | <b>0.147</b> | <b>0.126</b> | <b>0.092</b> | 0.038             |
| Mamba                     | 0.222        | 0.152        | 0.104        | 0.109             |
| <b>MOF/4 — first-last</b> |              |              |              |                   |
| GRU                       | 1.050        | 0.540        | 0.443        | 0.424             |
| LSTM                      | 0.935        | 0.681        | 0.392        | 0.698             |
| sLSTM                     | 0.848        | <b>0.414</b> | <b>0.258</b> | <b>0.262</b>      |
| Transformer               | 0.831        | 0.585        | 0.470        | 0.571             |
| Mamba                     | <b>0.798</b> | 0.693        | 0.575        | $2.0 \times 10^6$ |
| <b>MOF/6 — full obs</b>   |              |              |              |                   |
| GRU                       | —            | <b>0.067</b> | <b>0.054</b> | 0.030             |
| LSTM                      | —            | 0.080        | 0.056        | <b>0.029</b>      |
| sLSTM                     | <b>0.122</b> | 0.085        | 0.054        | 0.048             |
| Transformer               | —            | —            | 0.057        | 0.034             |
| Mamba                     | —            | —            | 0.062        | 0.054             |
| <b>MOF/6 — first-last</b> |              |              |              |                   |
| GRU                       | —            | 0.130        | 0.120        | 0.092             |
| LSTM                      | —            | —            | 0.106        | <b>0.087</b>      |
| sLSTM                     | <b>0.351</b> | <b>0.130</b> | 0.181        | 0.115             |
| Transformer               | —            | —            | <b>0.087</b> | 0.098             |
| Mamba                     | —            | —            | 0.114        | 0.132             |

**Table 6:** Architecture sweep — glycolysis — NRMSE (median).

| Model                       | $n = 3$      | $n = 10$     | $n = 100$    | $n = 1000$        |
|-----------------------------|--------------|--------------|--------------|-------------------|
| <b>Glyc/8 — full obs</b>    |              |              |              |                   |
| GRU                         | 0.146        | 0.112        | <b>0.102</b> | <b>0.090</b>      |
| LSTM                        | 0.269        | 0.132        | 0.106        | $6.4 \times 10^6$ |
| sLSTM                       | 0.184        | 0.111        | —            | —                 |
| Transformer                 | <b>0.117</b> | 0.114        | 0.110        | 0.096             |
| Mamba                       | 0.200        | <b>0.108</b> | 0.104        | 0.133             |
| <b>Glyc/8 — first-last</b>  |              |              |              |                   |
| GRU                         | 0.250        | 0.485        | 0.679        | 0.867             |
| LSTM                        | 0.206        | <b>0.300</b> | 0.675        | 0.894             |
| sLSTM                       | 0.154        | 0.704        | —            | —                 |
| Transformer                 | 0.201        | 0.537        | 0.551        | <b>0.796</b>      |
| Mamba                       | <b>0.131</b> | 0.387        | <b>0.512</b> | 0.823             |
| <b>Glyc/12 — full obs</b>   |              |              |              |                   |
| GRU                         | 0.610        | 0.266        | 0.132        | <b>0.065</b>      |
| LSTM                        | 0.371        | <b>0.191</b> | 0.348        | 0.073             |
| sLSTM                       | 0.338        | 0.312        | 0.128        | 0.067             |
| Transformer                 | <b>0.222</b> | 0.221        | <b>0.106</b> | 0.065             |
| Mamba                       | 0.249        | 0.257        | 0.154        | 0.110             |
| <b>Glyc/12 — first-last</b> |              |              |              |                   |
| GRU                         | <b>0.309</b> | 0.376        | 0.564        | 0.455             |
| LSTM                        | 0.384        | <b>0.319</b> | 0.525        | <b>0.369</b>      |
| sLSTM                       | 0.347        | 0.793        | 0.576        | 0.436             |
| Transformer                 | 0.396        | 0.359        | 0.463        | 0.441             |
| Mamba                       | 0.333        | 0.455        | <b>0.320</b> | 0.756             |